

Preferred educational requirements

Diploma/BSc/BE/BTech in Electronics & Communications or Electrical & Electronics

will be shown. Some designers perfect themselves in digital or analogue designs while others work on the routing layout. They make sure that the final design works as expected and meets industry requirements, before it is sent to the manufacturing facility for PCB fabrication.

EFY is opening training centres all over India where hands-on PCB designing is a favourite course among the engineering students and trainees from the industry. Mentioning about the 3-day introductory course, Ramesh Chopra, editor of EFY magazine and executive chairman of EFY Enterprises says, "Our objective is to introduce the students to PCB designing by teaching them the fundamentals using an open source software like gEDA. Since the trainees are from electronics field, they learn the skills quickly. Of course, they need to practice thereafter if they wish to become experts in the field."

Usually, each PCB designer works as part of a team whose members are responsible for different portions of the design. With large, complex boards, several designers can be assigned to each board and each person made responsible for a different set of functions. Even though a designer works independently, the team usually communicates through frequent meetings and thorough documentation. Therefore each designer is expected to communicate ideas verbally and explain

Skills expected

Freshers	Experienced
Technical skills	
1. Strong fundamentals in basic electronics 2. Basic understanding of electronic components and their package information 3. Basic understanding of PCB design tools such as Proteus, Eagle, Altium and ORCAD 4. Good aptitude	1. EDA tool knowledge 2. Exposure to manufacturing and PCB materials 3. Preference to CID qualification (certified interconnect designer from IPC) 4. Should be strong in transmission line theory. 5. Should be capable of designing different types of PCBs (designing for RF communication is different from designing for power and control applications) 6. Capability of designing more than two-layer PCBs 7. PCB troubleshooting in case of any track/via shortage or open
Soft skills	
1. Continuous learning drive 2. Communication skills	

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the PCB design in detail as well.

Near future

Though there has been a steady progress in the level of engagement between academia and industry, still there is a lot to be done suggest industry experts. Industries need to come forward and set up labs (with respect to DSP, FPGA, wireless technologies, EDA tools) for engineering colleges, which requires investment both in terms of engineering resources as well material (cost of development kits, tools). Kumar says, "But, at the same time, engineering colleges should be willing to accommodate industry-sponsored courses as

well as labs in their curriculum, which at times might look like brand building tools for sponsoring companies with one-sided views."

Industry and educational institutes should work together to identify and train employable graduates in this field, as there is a tremendous need for such niche activity. "PCB designing is a stable industry as development never ceases and growth is assured at the rate of more than 20% if the performance is excellent and there is a proven track record," believes Sathish. ●

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